

RADHIKA DOTIHAL

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Software Engineer | C++, Java, Python | Performance Optimization | AWS & GCP

5+ years of experience building high performance systems, reducing processing time and improving scalability.

TECHNICAL SKILLS

Programming Languages: C++, Java, Python, C

Web Technologies: HTML5, CSS3, PHP, JavaScript, jQuery, JSON, XML, Document Object Model (DOM)

Databases: MySQL, MongoDB, Cassandra, DynamoDB

Cloud & Containers: GCP, Docker, AWS (Lambda Function), Kubernetes, Jenkins

APIs: REST

Framework: Bootstrap, Express.js, JUnit, Mocha-Chai, Flask, LangChain, CrewAI

AI/ML & LLM Systems: OpenAI, Claude, LLM Orchestration, RAG

Version Control: GitHub

Tools: VS Code, Eclipse, Jupyter Notebook, Jira, Confluence

PROFESSIONAL EXPERIENCE

Professional Development

Sep 2025 – Present

- Optimized dense matrix multiplication in C++ by implementing cache blocking, loop unrolling, SIMD AVX2 vectorization and multi-threading, achieving speedup over the naïve implementation from 0.2 GFLOPS to 1.2 GFLOPS and improving cache utilization and CPU parallelism.
- Developed an AI-powered resume analyzer using all-MiniLM-L6-v2 model, pdfplumber, Streamlit and Sentence Transformers which use embedding semantic similarity scoring to compute ATS match percentage between resume and job description.

Software Engineer – SS&C Technologies, Inc. Advent [C++, Python, MySQL, Linux, Docker, Jenkins]

July 2022 – Aug 2025

- Accelerated report generation via multithreading and concurrency in C++, reducing processing time by 98.8%
- Improved performance issues in portfolio reports using references and inline functions, enhancing throughput.
- Normalized database schema, reducing query time from 15 sec to 3 sec for reports having more than 500 portfolios.
- Engineered a Python automation script to validate environment configurations, ensuring reliable development setups.
- Crafted a Python comparison tool using python dictionaries to detect and log differences across multiple runs.

Volunteer – Global Health Impact Project [Python, Flask, React.js, SQLite]

May 2021 – Aug 2021

- Implemented backend for website 'GHI Forecasting Tool' used to measure pharmaceutical product's impact on global health to increase access for medicines.
- Consolidated all the data into a single source of truth using Python for Global Health Impact Website 2021.

Software Engineer – LTI, India [Hyperion Infra, Oracle EPM, Java, MySQL, HTML, CSS, JavaScript]

Aug 2018 – Nov, 2020

- Orchestrated ETL migration across UAT, DEV, and QA using Informatica PowerCenter, streamlining data workflows.
- Led OBIEE repository migrations with Oracle BI Admin, Enterprise Manager, and Data Integrator.
- Developed an interactive website (HTML/CSS/JS) enabling users to query bus schedules, select seats, & view past bookings.
- Engineered a Java-based banking system with OOP inheritance for bank accounts, deposits and withdrawals.
- Outlined Online Banking Test Driven Model boiler plate using Java and performed Junit Testing.

ACADEMIC PROJECTS

Building & Performance Testing of a Highly Available Website using GCP Serverless Computing [JavaScript, Python, GCP]

- Deployed a website to track International Space Station and display location on Google Maps using JavaScript by utilizing features such as google cloud function, google cloud SQL and CRON job scheduler.
- Analyzed website performance by tracking the number of requests, latency and container count using Apache Benchmark tool.

Containers & Mini Docker [Python, Docker, Sysbench, QEMU, GCP]

- Implemented a Docker like tool called Mini Docker in Python to create container instances with namespace & control groups.
- Evaluated the performance of Docker container and QEMU using sysbench tool by running CPU and File I/O test modes.

Smart Home Automation using Alexa [Python, flask, AWS, XML, JSON]

- Instituted functionalities such as turning devices ON/OFF, adjusting brightness, setting schedules, creating and controlling device zones and providing security.
- Trained Alexa with intents and utterances in JSON using AWS Lambda function, maintained and tracked the database in xml.

Study of Software Defined Networking Techniques [Python, Google Cloud Platform, Kubernetes MiniKube]

- Investigated the performance of “of_tutorial” controller, MAC learning controller and MAC learning controller with open flow rules using Mininet, OpenFlow switches and POX.

Implementation of Design Patterns [Java]

- Designed student course registration system to check if the student satisfies the course and foreign language requirements using Template Method Design Pattern.
- Developed an application for users to select the topping of their own choice and display total price and calories using java applet and Decorator Design Pattern.
- Performed Nondeterministic Finite Automaton (NFA) to Deterministic Finite Automata (DFA) conversion to process input and displayed output as accepted or rejected using State Design Pattern.
- Performed arithmetic operation on spreadsheet using observer design pattern.

Mini Shell [C]

- Developed a shell (console), including system calls like fork, exec, wait, pipe to let the user run the commands with arguments, input and output file redirection and multiple commands separated by pipes.

Sales Forecasting using Time Series Data Analysis [Python]

- Forecasted the sales of an e-commerce business by building an ARIMA model and calculated the model's accuracy using RMSE.

Sales Analysis [Hadoop, Java]

- Analyzed the big data and visualized using pie charts and bar graphs using K-Means’ algorithm and Map Reduce.

Eatery Order [JavaScript, Express.js, REST, DOM, MongoDB, JSON, Mocha-chai]

- Established eateries backend using Test Driven Development approach to retrieve list of all eateries of a particular cuisine, displayed category for the specified eatery and tracked all menu items for a specified eatery and category by using map object in JavaScript.
- Processed data using the make factory method and used Mocha framework and Chai external Library for test cases.
- Designed the backend in MongoDB by setting up RESTful API routes and HTTP client to send requests.
- Incorporated web services to design a web page for the users to input cuisine and dishes of their interest from the drop-down.

EDUCATION

Master of Science, Computer Science State @ University of New York at Binghamton, New York

Feb 2021 – May 2022

Bachelor of Engineering, Information Technology @ University of Pune, India

Aug 2014 – May 2018

ACHIEVEMENTS & PUBLISHPMENT

- Presented and published research paper titled “[*Smart Homes Using Alexa and Power Line Communication in IoT*](#) | [*SpringerLink*](#)” at an annual ICCCNT conference in Springer. **April,2018**
- Completed online certification in AI Agents, ChatBot Development and LLM fundamentals through SimpliLearn **May, 2026**